FACULTY OF INFORMATION TECHNOLOGY CTU IN PRAGUE

Assignment of bachelor's thesis

Title:	Detection and Classification of Defects in Oak Logs from CT Scans
Student:	Oliver Morgan
Supervisor:	Ing. David Kramný
Study program:	Informatics
Branch / specialization:	Artificial Intelligence 2021
Department:	Department of Applied Mathematics
Validity:	until the end of summer semester 2026/2027

Electronically approved by Ing. Magda Friedjungová, Ph.D. on 2 February 2026 in Prague.

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Instructions

The aim of the thesis is the detection and classification of defects in oak logs scanned using computed tomography. The thesis will employ image processing methods in accordance with relevant standards specifying wood defects (ČSN ISO 1310, ČSN ISO 1311, etc.). The detection and classification of defects will serve as a basis for optimising the cutting strategy with the goal of maximising material utilisation and economic benefit. An unannotated dataset provided by the Czech University of Life Sciences Prague (CZU) is provided, containing CT scans of 40 oak logs with a length of 1 m and a vertical resolution of 1 mm. Expert consultation may be used for the dataset provided by CZU. For the purposes of machine learning, the student may search for and use existing annotated datasets.

Thesis tasks:
1. Conduct a literature review of methods for detecting and classifying defects in wood and related materials.
2. Define suitable defect classes for machine processing based on relevant standards and available data.
3. Design a methodology for efficient data preparation for further processing (creation/acquisition of annotations, etc.).
4. Design an algorithm for defect detection and classification using image processing methods.
5. Verify the results of the proposed algorithm and evaluate its performance.
6. Visualise the results of the algorithm on the image data.

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